

GAME_DESIGN.md — Mechanics, Economy, Progression, UX

Scope: how the game works. Learning/teaching rules live in EDUCATION.md; child personalization in CHARACTERS.md; code locations in DEVELOPMENT_RULES.md. Numbers below reflect the current implementation (`index.html` , v1.1.0) and are the baseline to preserve unless a documented change is approved.

Core loop

Pick profile → play the **Daily Mission** (the one featured action on the landing screen) → earn XP + coins → keep the streak → request rewards in the Store → parent approves → repeat daily.

Currencies (three, strictly separated)

Currency	Earned by	Spent on	Rules
Coins	Effort and completed missions	Store rewards, only via parent-approved request	Never deducted at request time; deducted only on approval after a balance re-check
XP	All learning activity	Nothing — progression only	Never spendable, never deducted
Adventure Keys	Streak milestones only: days 3, 7, 14, 30	Gate premium rewards (with coins)	Never from a single task; one key per milestone, tracked via <code>lastKeyMilestone</code>

Wrong answers never reduce XP, coins, or streaks (EDUCATION.md rule).

Levels & XP math (current values)

- Start: level 1, `xpMax` 100. Level-up: `xp -= xpMax; lvl++; xpMax += 50` . (Known inconsistency: `markMissionDone` in the parent dashboard uses `xpMax × 1.15` — unify to `+50` ; tracked in FUTURE_IDEAS.md → Now.)

- **Daily Mission (20 questions, `QCOUNT=20`):**
 - 8 XP per correct answer (`XP_PER_Q`), + mission base 140 XP,
 - accuracy bonus `round(accuracy/3)` .
 - Coins: `35 + round(accuracy/5)` .
 - **Weekend Double XP** is real (×2 on Sat/Sun, applied to XP only).
 - **Team bonus** ×1.15 XP if the sibling also completed their daily mission that day (friendly, optional cooperation — never head-to-head comparison, per CHARACTERS.md).

Streaks

- Completing the Daily Mission bumps the streak once per day.
- **Weekend freeze:** the streak only requires weekdays — playing Friday keeps it alive through Monday (`lastRequiredDay` walks back over Sat/Sun).
- Milestones 3/7/14/30 award an Adventure Key with celebration FX + sound.
- Streaks are never reduced as punishment; parent can adjust manually.

Missions & modes (current)

Mode	What it is	Notes
Daily Mission	Adaptive 20-question run, per-child themed	The featured daily action; engine below
Battle Arena	Turn-based sibling duel	Quick Duel (to 5), Rumble (to 8), Boss Clash (to 12); condensed 3-tier Brain Boost
Money Lab	Savings goal tracker + money challenges	Coins reward; real-world money math
Guided Mission	Briefing → Strategy → Solve → Explain → Check	Teaches method selection and self-checking
Equation Escape Room	Solve equations by choosing legal algebra "moves"	Locks open per solved equation
Grant Prize	Parent-PIN-gated instant	

chest	coin grants	For off-app achievements
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Mission theming is per child (Alex: battle/champion quests; Katya: detective mysteries) — theme content rules in CHARACTERS.md → Design guidance.

Adaptive difficulty (current engine — keep this behavior)

Per-child, per-topic mastery is recorded on every answer. For each topic:

- **Selection weight:** unexplored (<4 seen) = 3; accuracy <60% = 4 (struggling → more practice); 60–85% = 2; ≥85% = 1 (mastered → occasional review).
- **Difficulty tier (1–3):** <60% or unexplored = 1; 60–85% = 2; ≥85% = 3.
- The 20-question sequence is weight-proportional, shuffled, with no topic appearing 3× in a row; recently seen question text (buffer of 60) is avoided so questions rarely repeat. Every generator produces fresh random numbers.
- “Weak/strong subjects” insights surface topics <70% (weak) and ≥85% (strong) once seen ≥3 times.

Achievements & badges

- Six hex badges exist visually (First Win, Streak 7, Math Genius, Early Bird, Team Player, Perfectionist) — currently decorative/aspirational.
- **Strategy badges** unlock through Brain Boost strategy usage.
- Future achievement ideas go to FUTURE_IDEAS.md, not here.

Store & reward approval (exact flow — do not weaken)

1. Child taps **Request Reward** on an affordable item. Kids can never redeem directly.
2. Request created; item shows *Pending Parent Approval*. **Nothing is deducted or escrowed at request time.**
3. Duplicate pending requests for the same item are blocked; can’t request an unaffordable item. Child sees their own request history.
4. Parent dashboard lists pending requests (child, reward, cost, date).
5. **Approve** re-checks balance at that moment, then deducts coins (and keys for premium). If balance no longer covers it, approval is refused gracefully.

6. **Decline** deducts nothing.

7. Full history persists.

Catalog: rarity tiers Common → Rare → Epic → Legendary → Mythic → Ultra, prices editable by parent; rewards can be disabled (hidden from kids). Premium rewards require BOTH currencies: \$5 = 300 🌐 +1 🔑, \$10 gift card = 500 🌐 +2 🔑, \$15 game/app = 700 🌐 +3 🔑. No reward may ever be a "skip practice/chore/day".

Parent dashboard (PIN-gated)

Insights (weak/strong topics), activity log, Rewards Manager, manual adjustments (coins/XP/streak/keys, mark mission done), weekly summary per child, PIN change (default 1234 — change in production), sound mute toggle, app version + update check, and a guarded full progress reset.

Accessibility & UX principles

- **Tablet-first** (Alex: iPad; Katya: Lenovo Android tablet), works on phone and desktop. Touch-first: large targets, no hover-only interactions, no horizontal scroll, safe orientation changes, working numeric/decimal keyboards (including a sign-toggle for negatives).
- One clear featured action on landing (Daily Mission) — never trap the child; Switch button always returns to the profile picker.
- Animations use CSS transforms/opacity and honor `prefers-reduced-motion`.
- Sound: short soft Web Audio cues (task, coin, streak, key, request, approve); global persisted mute. No harsh failure sounds — mistakes are quiet, wins are celebrated (confetti bursts scale with achievement).
- Visual identity: dark neon-arcade, Orbitron display font, per-child accent colors and hero photos. Preserve this look during any migration.